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# Predicting fall risk through step width variability at increased gait speed in community dwelling older adults

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**We aimed to investigate the relationship of fall risk with spatiotemporal gait variables and variability in community-dwelling older adults and identify the key predictors of fall risk. A total of 303 older adults were classified into the fall-risk and non-fall-risk groups based on their Short Physical Performance Battery scores. The stance phase time, double stance time, stride length, step width, velocity, cadence, and coefficient of variation (CV) for each variable were measured on a treadmill at the preferred speed and 20% increased walking speed. The results showed that the fall-risk group had a significantly increased stance phase and double stance time, while the stride length, velocity, and cadence were significantly reduced compared to the non-fall-risk group at both speeds. Additionally, the CV for most variables was significantly higher in the fall-risk group. Notably, the step width variability was higher in the fall-risk group under both speed conditions and showed greater discriminative power at increased walking speed. Thus, gait variability, particularly step width variability, is a crucial predictor of fall risk. The increased variability observed in the fall-risk group at higher walking speed suggests that it could be a useful early indicator for predicting fall risk.**

**Keywords** Fall risk, Gait variability, Spatiotemporal gait parameters, Short physical performance battery, Step width variability

With the rapid aging of the society, falling has gained prominence as one of the most common and serious issues affecting older adults. Falls are not only a significant cause of physical injury in older adults but also one of the primary factors leading to a loss of independence[1], which greatly diminishes the quality of life. In the long term, falls can also contribute to increased healthcare costs[2]. Injuries resulting from falls are particularly difficult to recover from, and the subsequent complications can drastically reduce an individual's range of activity, potentially leading to secondary health deterioration.

In community-dwelling older adults, reduced gait speed[3], impaired balance ability[4], and decreased lower limb strength have been associated with an increased risk of falls[5]. Therefore, the fall risk in older adults can be assessed using measures such as gait speed, balance ability (e.g., postural sway), and functional tasks, such as the sit-to-stand test and lower limb strength evaluations. In fact, the Short Physical Performance Battery (SPPB) is an objective tool for measuring the gait speed, balance, and lower extremity strength in older adults, and has been shown to be a quick and useful tool for screening community-dwelling older adults at risk of falls[6,7]. Furthermore, Most falls occur during movement[8]; therefore, the current fall prevention guidelines for older adults recommend gait assessment[9]. The SPPB is a widely used fall prediction tool and was recently recommended as a key risk assessment tool for fall prevention and management of older adult health in the World Falls Guidelines, developed by the World Falls Task Force consisting of 96 multidisciplinary experts representing 36 scientific and academic societies from 39 countries[10]. A score of 9 or below out of 12 on the SPPB is considered indicative of an increased risk of falling[11].

Previous studies have reported that among the spatiotemporal gait parameters, slower walking speed[12], longer stance phase, and increased double stance phase[13] are associated with falls in older adults. Spatiotemporal variables are useful for distinguishing prospective fallers from non-fallers among older adults[13]. In older adults with normal walking speed, both high and low stride variability are associated with a history of falls[14]. Additionally, fear of falling is closely related to increased gait variability[15]. Gait variability refers to the stride-to-

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stride variation of spatiotemporal gait parameters such as stride length, step width, and stance time, and thus reflects the consistency or stability of gait parameters[16], and plays an important role in assessing the dynamic balance during walking in healthy older adults[17]. In addition, an increased variability in stride length, stance phase, and gait speed is associated with mobility disability and falls in older adults[18–20]. In contrast, low gait variability is indicative of greater gait stability[16,21]. Among the various gait variabilities, step length and velocity variability primarily reflect anteroposterior control, whereas step width variability is more sensitive to lateral instability[22], a factor known to contribute to loss of balance and falls. previous study have highlighted step width variability as a meaningful descriptor of locomotor control, showing stronger associations with balance and fall risk compared to other gait variability measures such as step length[23]. Therefore, variability of gait parameters can be used as key indicators for assessing the risk of falls. Poujol et al.[24] recommended that spatiotemporal gait parameters, as well as gait variability in older adults, should be analyzed at faster walking speeds than the preferred speed (PS). As such, previous studies have assessed fall risk using the averaged values of gait parameters at PS, while the relationship between the variability of these gait parameters and fall risk remains unclear. *In addition, research examining gait parameters and variability at faster walking speeds is limited, and the differences in gait characteristics between PS and fast walking conditions have not been fully identified. A previous study compared the gait variability between preferred walking speed and a 20% faster walking speed and reported that the coefficient of variation (CV) increased under the faster speed condition[25]. Accordingly, identifying the variability of gait parameters at faster walking speeds may serve as a novel indicator for predicting fall risk in older adults.*

This study aimed to identify spatiotemporal gait parameters and variability measures that can predict or distinguish fall risk in community-dwelling older adults. Participants were classified into fall risk and non-fall-risk groups based on SPPB scores, and gait characteristics were analyzed at preferred speed and with a 20% increase in walking speed. We hypothesized that gait variability in spatiotemporal parameters would significantly differ between fall-risk and non-fall-risk groups at both preferred speed and 20% increased speed and serve as a stronger predictor of fall risk than average gait values.

## Methods

### Design and ethical considerations

Between December 2023 and March 2024, we enrolled 303 older adults and assessed the variability in their gait parameters at the PS and 20% faster walking speed. Prior to the start of the study, the participants were thoroughly briefed on the study procedures and safety measures, and signed informed consent was obtained from each participant.

### Participants and procedures

This study recruited 303 older adults aged 65–92 years from community centers using posters and telephone interviews. The eligibility criteria were as follows: (1) aged ≥ 65 years and capable of performing daily activities independently or with aid; (2) without prior orthopedic surgery involving the lower limbs, no current neurological conditions, such as cerebral infarction; and (3) no surgeries performed on the musculoskeletal system of the lower limbs within the past 6 months. The exclusion criteria were as follows: a score below 24 on the Korean version of the Mini-Mental State Examination (K-MMSE), a diagnosis of lower extremity musculoskeletal disorders that significantly affect gait (e.g., symptomatic knee or hip osteoarthritis, history of lower limb fracture), and difficulty walking on a treadmill or inability to complete the required gait assessments. Initially, 364 individuals were considered; however, 14 did not satisfy the inclusion criteria and 47 were unable to complete the required measurements owing to discomfort or difficulty in walking on the treadmill.

After an initial assessment of the general characteristics and cognitive functions, the fall risk of all participants was evaluated. This was followed by measurement of the gait parameters and variability while the participants walked on a treadmill. To ascertain their preferred walking speed, each participant walked a 14-m stretch before treadmill use. To reduce measurement errors pertaining to the average preferred walking speed due to a 20% acceleration factor, only the time taken to walk the central 10 m was considered, excluding the first and last 2 m. Subsequently, the participants were adapted to the treadmill using the walking speed established from the overground trials. Any discrepancies in speed during the treadmill adaptation led to minor adjustments in the PS by ± 0.1 km/h. Once the PS was set, the participants took a 2-min rest before beginning their treadmill walk at the PS. Next, the participants walked for 90 s at both their PS and 20% increased speed, with a 1-min rest between each trial. The gait variables and variability were measured for approximately 60 s, starting at 20 s after the participants began to walk on the treadmill.

All participants were randomly assigned to one of the two speed sequences (20% faster walking speed > PS or PS > 20% faster walking speed). The order in which the participants walked was assigned randomly using a random number table. To ensure fairness and objectivity, the randomization process was performed by a researcher who was not directly involved in the study. Data collection was performed in university laboratories and community centers. The outcome assessors were blinded to the group allocation.

### Measurement

*Fall risk*

The fall risk was assessed using the SPPB, which is used for screening the fall risk in older adults[6]. The inter-rater reliability of the SPPB in community-based older adults has been reported to be excellent (Intraclass Correlation Coefficient [ICC] = 0.81–0.91)[26]. The SPPB (ranging 0–12 points) evaluates the physical performance using three components: a balance test, 4-m walk, and timed chair stand test. In the balance test, the participants were required to maintain three different positions-feet together, semi-tandem, and tandem—for 10 s each. The 4-m walk test was performed at the participants' normal walking speed. For the chair stand test, the participants first performed five practice stands, and then their time to complete five consecutive stands from sit to stand was recorded in seconds. As an SPPB score of ≤ 9 is associated with an increased fall risk[11], this threshold was used to indicate the fall risk in this study. Therefore, we defined a score of ≤ 9 as indicating a fall risk and a score of ≥ 10 as indicating no fall risk.

*Gait parameters and variability*
The gait parameters and variability were measured using the Zebris FDM–THM–S treadmill system (Zebris Medical GmbH, Max–Eyth–Weg 43, D–88316, Isny, Germany). The system consists of a capacitive pressure platform integrated into a treadmill. The pressure platform has a sensing area of 108.4 × 47.4 cm, which is incorporated with 7,168 sensors, each approximately 0.85 × 0.85 cm in size. The treadmill has a contact surface area of 150 × 50 cm, and the belt speed can be adjusted between 0.2 and 22 km/h in 0.1 km/h increments. The contact surface of the treadmill remains horizontal (0% inclination) throughout the tests. The participants underwent an adaptation session on the treadmill, during which they were briefed on the safety procedures and given approximately 2 min to familiarize themselves with walking at the PS. Most gait parameters, including stride length (ICC = 0.94), step width (ICC = 0.97), and double stance phase time (ICC = 0.86), have shown very high reliability during treadmill walking in community-dwelling older people[27]. A common measure of the magnitude of variability is the coefficient of variation (CV).

In this study, we have reported the results using standard deviation (SD) because the SD and CV of the gait variables are highly correlated[18]. The variability of the measured gait parameters was expressed as a percentage and was calculated by dividing the standard deviation of each gait parameter by the mean (standard deviation/mean × 100). CV is a very important tool for quantitatively assessing and understanding gait stability and quality. In fact, gait variability measured by CV has been identified as the most clinically relevant digital biomarker for assessing gait impairment[28]. Thus, gait variability can be used to assess the fall risk in community-dwelling older adults[18].

To reliably assess gait variability, it is recommended to walk at least 50 steps or for > 1 min[29,30]. The gait parameters obtained in this study were spatiotemporal parameters, including the following:

- Stance phase time: part of the gait cycle. It refers to the duration when the foot is in contact with the ground and is expressed as a percentage (%).
- Double-stance phase time: part of the gait cycle. It refers to the period when both feet are in contact with the ground simultaneously and is expressed as a percentage (%).
- Stride length: distance between the two heel contacts on the same side of the body, expressed in cm.
- Step width: lateral (side-to-side) distance between the feet during walking. It is measured from the midpoint of one foot to the midpoint of the other foot and is expressed in cm.
- Velocity: refers to the walking speed and is typically measured in kilometers per hour (km/hour).
- Cadence: the number of steps taken per minute, expressed as steps per minute (step/min).

## Statistical analysis
SPSS 26.0 software (IBM, Armonk, NY, USA) was used for statistical analyses. Frequency and descriptive statistics were used to describe the general characteristics of the participants.

The participants were divided into the fall-risk and non-fall-risk groups based on their fall risk. The differences in gait parameters and variability between the two groups at the PS and 20% faster speed were compared and analyzed using the independent t-test. To investigate the effect of sex on gait variables and variability in the groups, ANCOVA was performed with age as a covariate. To identify which gait parameters and variability were significantly correlated with the SPPB score in the entire cohort, multiple regression analysis was performed at both the PS and 20% faster walking speed. Additionally, discriminant analysis was conducted to identify the gait parameters and variability that could distinguish between the fall-risk and non-fall-risk groups. All data are expressed as mean ± standard deviation.

# Results
The fall-risk and non-fall-risk groups comprised 136 (mean age: 80.99 years) and 167 (mean age: 78.87 years) participants, respectively. Table 1 summarizes the characteristics of the participants.

## Comparison of the gait parameters and variability at the PS and 20% faster speed between the fall-risk and non-fall-risk groups
On comparing the gait parameters and variability at the PS and 20% increased speed between the fall-risk and non-fall-risk groups, no significant differences were noted in the double stance and step width at the PS and step width at 20% increased speed. Compared to the non-fall-risk group, the fall-risk group showed significantly increased stance phase time and double stance time ($p < 0.01$) at both the PS and 20% increased speed, as well as a significantly shorter stride length ($p < 0.01$), slower velocity ($p < 0.01$), and lower cadence ($p < 0.01$). The variability in gait parameters, excluding the CV of the double-stance time at the PS, was significantly increased for all other CVs in the fall-risk group. In comparisons between the two groups, age affected the stance phase time at PS, and stance phase time %, stride length, velocity, and cadence at 20% increased speed (Table 2).

## Multivariable regression analysis to identify the factors influencing the SPPB score based on age, sex, and gait variables and variability in older adults
At the PS, age (β = -0.086, $p = 0.034$), double stance time (β = 0.331, $p = 0.014$), stride length (β = 0.925, $p = 0.002$), step width CV (β = -0.265, $p < 0.001$), velocity (β = -0.810, $p = 0.009$), cadence (β = 0.505, $p < 0.001$), and cadence CV (β = -0.224, $p = 0.001$) were identified as the significant factors influencing the SPPB score. In particular, the step width CV and cadence demonstrated very low p-values, indicating a significant effect on walking speed.

| | Fall risk group ≤9 SPPB score ($n$=136) | Non-fall risk group ≥10 SPPB score ($n$=183) | t | $p$ |
|---|---|---|---|---|
| Sex (male/female) | 73 (45.6) / 86 (54.4) | 62 (43.7) / 80 (56.3) | | |
| Age (years) | 82.56 ± 6.03 | 78.24 ± 6.01 | 6.335 | < 0.001 |
| Weight (Kg) | 60.44 ± 9.79 | 61.35 ± 8.88 | -0.866 | 0.387 |
| Height (cm) | 157.57 ± 7.56 | 159.10 ± 8.31 | -1.684 | 0.093 |
| SPPB (score) | 7.62 ± 1.84 | 11.36 ± 0.75 | -24.731 | < 0.001 |
| K-MMSE (score) | 25.17 ± 2.74 | 26.54 ± 2.63 | -4.493 | < 0.001 |
| Marital status | | | | |
| Married | 43 (31.6) | 102 (61.1) | | |
| Divorce | 11 (8.1) | 11 (6.6) | | |
| Widowed | 77 (56.6) | 49 (29.3) | | |
| Never married | 5 (3.7) | 5 (3.0) | | |
| Current health condition | | | | |
| Very good | 40 (30.1) | 79 (47.3) | | |
| Average | 49 (36) | 58 (34.7) | | |
| Bad | 46 (33.8) | 30 (18.0) | | |
| Number of exercises in a week | | | | |
| 3 or more times | 90 (66.2) | 152 (91.0) | | |
| 1–2 times | 13 (9.6) | 8 (4.8) | | |
| no | 33 (24.3) | 7 (4.2) | | |
| Taking medication (multiple response) | | | | |
| Hypertension | 74 (54.4) | 87 (52.1) | | |
| Diabetes | 43 (31.6) | 39 (23.4) | | |
| Musculoskeletal disease | 5 (3.7) | 4 (2.4) | | |
| Cardiac disease | 6 (4.4) | 10 (6.0) | | |
| Digestive system disease | 4 (2.9) | 5 (3.0) | | |
| Ect (prostate, pain killer…) | 59 (43.4) | 64 (38.3) | | |
| No | 37 (27.4) | 50 (29.9) | | |

**Table 1**. General characteristics of participants according to SPPB score. [a]number of person (%) SPPB, short physical performance battery; K-MMSE, Korean Mini-Mental State Examination.

When the walking speed was increased by 20%, the double stance time (β = 0.345, $p$ = 0.016), stride length (β = 1.248, $p$ < 0.001), step width CV (β = 0.300, $p$ < 0.001), velocity (β = -1.279, $p$ < 0.001), and cadence (β = 0.646, $p$ < 0.001) showed a significant association with the SPPB score, with the impact of the step width CV increasing at the 20% increased speed. Under both conditions, the stride length, step width CV, velocity, and cadence demonstrated a significant association with the SPPB score (Table 3).

On performing discriminant analysis, most gait parameters and variability showed statistically significant differences at the PS and 20% increased speed. At the PS, the stance phase time (Wilks' Lambda = 0.839, $p$ < 0.001), double stance time (Wilks' Lambda = 0.828, $p$ < 0.001), stride length (Wilks' Lambda = 0.872, $p$ < 0.001), velocity (Wilks' Lambda = 0.811, $p$ < 0.001), and cadence (Wilks' Lambda = 0.921, $p$ < 0.001) differed significantly. In terms of the CV, all variables except the double-stance CV demonstrated significant differences. Notably, the step width CV (Wilks' Lambda = 0.759, $p$ < 0.001) exhibited a high discriminative power.

At 20% increased gait speed, the double stance time, stride length, velocity, cadence, and CV of the step width and cadence demonstrated significant differences, with the step width CV showing a particularly high discriminative power (Wilks' Lambda = 0.697, $p$ < 0.001) (Table 4).

## Discussion

We examined the differences in gait variables and variability at both the PS and 20% increased speed between the fall-risk (older adults with low SPPB scores) and non-fall-risk groups (older adults with high SPPB scores). We also identified the gait variables and variability that could predict the fall risk in older adults and distinguish individuals at high risk of falling. Treadmill-based gait analysis allows for the collection of multiple consecutive gait cycles, and has been reported to simplify the measurement of gait speed[31]. Treadmill assessment allows researchers to precisely control the gait speed and monitor the gait at multiple predefined speeds[31,32]. This is particularly advantageous when studying populations with limited mobility. Additionally, it ensures that all participants are tested under the same conditions, reducing variability due to external factors. These findings support the reliability and objectiveness of the gait analysis data obtained in this study[33,34].

The results showed that the fall-risk group had a significantly increased stance phase time and double stance time at both the PS and 20% increased walking speed compared to the non-fall-risk group. In contrast, the stride length, velocity, and cadence were significantly reduced in the fall-risk group. Additionally, most variables showed significantly increased variability in the fall-risk group. These findings are consistent with the tendency

| Variables (unit) | Fall risk group | Non-fall risk group | t | *p*-value | Difference (95% CI) |
|---|---|---|---|---|---|
| **Preferred gait speed** | | | | | |
| Stance phase (%) | 69.92 ± 5.57 | 66.18 ± 2.75 | 7.597 | < 0.001 | 3.736 (2.768 ~ 4.704) |
| Stance phase CV (%) | 3.84 ± 2.85 | 2.86 ± 1.73 | 3.645 | < 0.001 | 0.974 (0.449 ~ 1.499) |
| Double stance (%) | 38.93 ± 9.86 | 32.01 ± 4.98 | 7.911 | < 0.001 | 6.919 (5.198 ~ 8.641) |
| Double stance CV (%) | 8.53 ± 8.41 | 7.01 ± 4.83 | 1.848 | 0.066 | 1.426 (0.092 ~ 2.945) |
| Stride length (cm) | 57.50 ± 19.73 | 73.72 ± 22.22 | -6.642 | < 0.001 | -16.220 (-21.026 ~ 11.414) |
| Stride length CV (%) | 9.30 ± 7.55 | 5.81 ± 3.81 | 5.178 | < 0.001 | 3.488 (2.162 ~ 4.813) |
| Step width (cm) | 11.27 ± 3.95 | 10.78 ± 3.26 | 1.168 | 0.244 | 0.484 ( -0.331 ~ 1.300) |
| Step width CV (%) | 25.53 ± 15.50 | 13.24 ± 4.61 | 9.767 | < 0.001 | 12.334 (9.849 ~ 14.819) |
| Velocity (km/h) | 1.82 ± 0.72 | 2.58 ± 0.84 | -8.384 | < 0.001 | -0.764 (-0.944 ~ -0.585) |
| Velocity CV (%) | 10.04 ± 8.53 | 5.36 ± 3.71 | 6.381 | < 0.001 | 4.680 (3.237 ~ 6.123) |
| Cadence (step/min) | 106.44 ± 22.43 | 117.78 ± 16.47 | -5.071 | < 0.001 | -11.346 ( -15.750 ~ -6.943) |
| Cadence CV (%) | 5.65 ± 4.68 | 3.31 ± 2.18 | 5.730 | < 0.001 | 2.338 (1.535 ~ 3.140) |
| **20% increased walking speed** | | | | | |
| Stance phase (%) | 68.67 ± 4.63 | 65.27 ± 2.70 | 7.935 | < 0.001 | 3.399 (2.556 ~ 4.242) |
| Stance phase CV (%) | 3.42 ± 2.20 | 2.58 ± 1.39 | 3.861 | < 0.001 | 0.808 (0.396 ~ 1.220) |
| Double stance (%) | 36.71 ± 8.96 | 30.19 ± 4.94 | 8.005 | < 0.001 | 6.525 (4.921 ~ 8.129) |
| Double stance CV (%) | 7.71 ± 4.52 | 6.29 ± 3.08 | 3.234 | 0.001 | 1.422 (0.556 ~ 2.288) |
| Stride length (cm) | 63.59 ± 21.06 | 83.77 ± 24.34 | -7.592 | < 0.001 | -20.177 (-25.408 ~ 14.947) |
| Stride length CV (%) | 7.07 ± 6.07 | 4.57 ± 3.06 | 5.790 | < 0.001 | 3.128 (2.065 ~ 4.192) |
| Step width (cm) | 10.86 ± 3.67 | 10.80 ± 3.28 | 0.138 | 0.890 | 0.055 (-0.734 ~ 0.845) |
| Step width CV (%) | 27.22 ± 12.14 | 14.61 ± 4.95 | 11.410 | < 0.001 | 12.615 (10.439 ~ 14.791) |
| Velocity (km/h) | 2.13 ± 0.86 | 2.99 ± 0.99 | -7.901 | < 0.001 | -0.858 (-1.072 ~ 0.644) |
| Velocity CV (%) | 8.31 ± 6.76 | 4.40 ± 3.40 | 6.285 | < 0.001 | 3.802 (2.611 ~ 4.992) |
| Cadence (step/min) | 111.68 ± 21.69 | 119.20 ± 15.74 | -3.477 | < 0.001 | -7.513 (-11.767 ~ -3.260) |
| Cadence CV (%) | 4.76 ± 3.46 | 2.95 ± 2.15 | 5.40 | < 0.001 | 1.808 (1.166 ~ 2.451) |

**Table 2**. Comparison of gait variables and variability between fall risk and Non-Fall risk groups according to Changes in Gait Speed. CV: coefficient of variation, CI: confidence interval.

of individuals with a history of falls to exhibit longer stance and double stance phase durations[13]. A double stance plays a crucial role in maintaining balance and stability, and an increase in the double stance time indicates the need for more time to stabilize the gait, especially at slower walking speeds[35]. Furthermore, falls are more strongly related to reduced walking speeds and shortened stride lengths than to age itself[13]. These results suggest that the fall-risk group had slower walking speed and prolonged stance phase time and double stance time to maintain walking stability compared to the non-fall-risk group. In particular, the study found that the variability in the double stance time was significantly increased only at the 20% increased speed in the fall-risk group. This may be due to the greater imbalance caused by walking at a faster speed than at the PS, as it requires more strategic adjustments to stabilize the gait[36]. Additionally, the step width variability was greater in the fall-risk group at both the PS and 20% increased walking speed. Step width variability reflects the amount of active control required for lateral stabilization during walking[22], indicating that the fall-risk group experienced reduced lateral stability during walking. In previous studies, stride length and velocity were reported to decrease with age, while the cadence increased[37]. Slower speed and longer stance phase may be related to impaired balance ability, which may increase the risk of falls in older individuals[38]. In our result, under the PS condition, older age was associated with increased stride length variability (CV) and cadence variability (CV). Furthermore, lower variability in stride length and cadence was related to higher SPPB scores, indicating that greater gait consistency is linked to better physical function in older adults. In addition, in PS, SPPB scores decreased with increasing age, step width CV, and cadence CV, and with decreasing step length and cadence. In contrast, under the 20% increased gait speed condition, higher double stance time, step width variability (CV), and gait velocity were associated with lower SPPB scores, while a decrease in cadence also corresponded with reduced physical performance. Additionally, when gait speed increased by 20%, stride length, stance phase time, double stance time, and velocity all increased, whereas cadence showed a decreasing trend. These findings suggest that changes in gait speed influence various spatiotemporal gait characteristics and may affect functional mobility in older adults. These results imply a correlation with increasing age, consistent with the results of previous studies. Therefore, we believe that it is more efficient to confirm changes in the gait variables at high walking speed.

*Guimaraes and Isaacs*[39] *reported that increased variability in step length and double support time was significantly associated with a higher risk of falls, whereas Brach et al.*[14] *found no significant relationship between mean step length or stance time and falls. In the present study, we confirmed that the step-width CV predicted fall risk more accurately than mean stride length and stride length CV under both PS and 20% increased speed conditions. These findings suggest that measures of variability may serve as more sensitive indicators of fall risk than*

| Independent variable | B | β | t | *p*-value | Adj.$R^2$ |
|---|---|---|---|---|---|
| **Preferred gait speed** | | | | | |
| Age (year) | -0.034 | -0.086 | -2.127 | 0.034 | 0.544 |
| Sex | 0.290 | 0.062 | 1.501 | 0.134 | |
| Stance phase time (%) | -0.008 | -0.016 | -0.135 | 0.893 | |
| Stance phase time CV (%) | 0.023 | 0.023 | 0.402 | 0.688 | |
| Double stance time (%) | -0.091 | -0.331 | -2.473 | 0.014 | |
| Double stance time CV (%) | 0.033 | 0.097 | 1.914 | 0.057 | |
| Stride length (cm) | 0.094 | 0.925 | 3.171 | 0.002 | |
| Stride length CV (%) | 0.063 | 0.168 | 1.952 | 0.052 | |
| Step width (cm) | 0.018 | 0.028 | 0.656 | 0.512 | |
| Step width CV (%) | -0.048 | -0.265 | -6.495 | < 0.001 | |
| Velocity (km/hour) | -2.121 | -0.810 | -2.639 | 0.009 | |
| Velocity CV (%) | -0.044 | -0.130 | -1.483 | 0.139 | |
| Cadence (step/min) | 0.057 | 0.505 | 3.434 | < 0.001 | |
| Cadence CV (%) | -0.138 | -0.224 | -3.259 | 0.001 | |
| **20% increased gait speed** | | | | | |
| Age (year) | -0.005 | -0.014 | -0.332 | 0.740 | 0.537 |
| Sex | 0.238 | 0.051 | 1.256 | 0.210 | |
| Stance phase time (%) | 0.025 | 0.045 | 0.334 | 0.739 | |
| Stance phase time CV (%) | 0.054 | 0.043 | 0.796 | 0.427 | |
| Double stance time (%) | -0.102 | -0.345 | -2.427 | 0.016 | |
| Double stance time CV (%) | -0.002 | -0.004 | -0.072 | 0.943 | |
| Stride length (cm) | 0.115 | 1.248 | 4.415 | < 0.001 | |
| Stride length CV (%) | -0.036 | -0.078 | -0.729 | 0.467 | |
| Step width (cm) | 0.030 | 0.046 | 1.090 | 0.277 | |
| Step width CV (%) | -0.060 | -0.300 | -7.137 | < 0.001 | |
| Velocity (km/hour) | -2.855 | -1.279 | -4.206 | < 0.001 | |
| Velocity CV (%) | -0.007 | -0.016 | -0.144 | 0.866 | |
| Cadence (step/min) | 0.078 | 0.646 | 5.063 | < 0.001 | |
| Cadence CV (%) | -0.073 | -0.094 | -1.632 | 0.104 | |

**Table 3**. Multivariable regression analysis to identify the factors influencing the SPPB score based on the age, sex, and gait variables and variability in older adults. CV: coefficient of variation.

| Variable | Wilks' Lambda | Chi-square | *p*-value |
|---|---|---|---|
| **Preferred gait speed** | | | |
| Stance phase (%) | 0.838 | 55.868 | < 0.001 |
| Stance phase CV (%) | 0.952 | 15.669 | < 0.001 |
| Step width (cm) | 0.994 | 0.891 | 0.169 |
| Step width CV (%) | 0.755 | 88.882 | < 0.001 |
| Velocity (km/h) | 0.797 | 71.856 | < 0.001 |
| Velocity CV (%) | 0.876 | 41.727 | < 0.001 |
| Cadence (step/min) | 0.922 | 25.859 | < 0.001 |
| Cadence CV (%) | 0.905 | 31.605 | < 0.001 |
| **20% increased gait speed** | | | |
| Stance phase (%) | 0.829 | 58.922 | < 0.001 |
| Stance phase CV (%) | 0.948 | 16.704 | < 0.001 |
| Step width (cm) | 0.999 | 0.205 | 0.651 |
| Step width CV (%) | 0.701 | 111.821 | < 0.001 |
| Velocity (km/h) | 0.820 | 62.553 | < 0.001 |
| Velocity CV (%) | 0.874 | 42.283 | < 0.001 |
| Cadence (step/min) | 0.957 | 13.716 | < 0.001 |
| Cadence CV (%) | 0.906 | 31.059 | < 0.001 |

**Table 4**. Discriminant analysis of gait variables and variability based on SPPB scores.

*mean values*. The findings of this study align was similar to previous studies that found that step width variability is a more meaningful indicator of gait stability than step length variability[23].

Previous biomechanical study shown that lateral foot placement is closely linked to the lateral position and velocity of the center of mass during walking, particularly during the single-limb support phase[40]. Mediolateral balance such lateral stability requires more active stabilization than anteroposterior balance, which is achieved primarily through lateral foot placement[22] and appropriate muscle control[41]. Thus, lateral stability during gait is maintained through the passive dynamics of the musculoskeletal system and the active control of the central nervous system[42,43]. A key mechanism that ensures lateral stability during walking is step width[44]. Step width variability reflects the degree of active control required to maintain this lateral stability[22], and instability in lateral foot placement during walking may indicate instability in active motor control. Instability in motor control can reduce balance ability, and balance impairments lead to greater alterations in gait patterns and an increased risk of falling during walking[45]. Therefore, the observed increase in step width CV at both PS and 20% faster speed suggests greater mediolateral instability, which may reflect not only impaired balance ability but also an increased risk of falls.

Discriminante analysis was used to determine the gait parameters and variability that could identify older adults at risk of falling. Under the PS condition, step width CV showed the lowest Wilks' Lambda value, although the difference compared to gait velocity was not significant. However, at 20% increased gait speed, Wilks' Lambda values for step width CV and velocity were 0.697 and 0.827, respectively, indicating a significant difference in the discriminatory ability between the two variables. When instability is perceived, step width tends to increase[22,46], and at increased walking speeds at PS, step width may increase further to ensure medial stability[22]. *This means that when the walking speed increases in the presence of instability, the step width variability may be increased. In the present study, gait speed consistently showed lower discriminative power than step width CV across all walking conditions. While gait speed is widely used as a basic assessment tool in clinical settings due to its simplicity and ease of measurement, its predictive validity for identifying fall risk in community-dwelling older adults has not yet been firmly established*[47]. *This suggests that although gait speed may be partially related to fall risk, it has limitations in capturing subtle changes in gait stability. Despite its clinical convenience, gait speed alone may not be sufficient for detecting fall risk, which is often associated with medial-lateral instability. Step width CV provides unique insights into gait variability and lateral stability that are not easily captured by gait speed alone.*

*Therefore, the authors suggest that step width CV measured under the condition of a 20% increased walking speed provides unique insights into gait variability and lateral stability—factors that are not easily captured by gait speed alone—and may serve as a more sensitive and meaningful indicator for detecting fall risk in older adults.*

These findings suggest that gait variability-particularly step width variability- can be used to distinguish between older adults at high and low risk of falling, especially under conditions of increased walking speed. Previous studies have reported that gait variability is a key factor for predicting the fall risk[48], and present study provides further supporting evidence. Notably, step width variability is a more meaningful indicator of gait control than speed and stride length variability[23]. Therefore, step width variability can be a useful indicator for predicting the fall risk among older adults.

Despite these findings, this study has some limitations. First, the data were collected from older adults living in the community, which may limit the generalizability of the results to other populations, particularly older adults residing in care facilities or other demographic groups.

Second, *this study was conducted under treadmill walking conditions, which may differ from natural overground walking. Since treadmill walking can constrain natural gait patterns, caution is needed when generalizing the results to real-world walking. Therefore, future studies should include validation analyses comparing treadmill and overground gait patterns to enhance the ecological validity of the findings.* Third, as this study did not evaluate actual fall experiences, a potential discrepancy may exist between those identified as being at risk of falling based on SPPB scores and those who may ultimately experience a fall. To overcome this limitation, future research should include longitudinal data on real fall incidents to further validate the predictive accuracy of the SPPB. Although the SPPB score was used to assess fall risk in this study, the SPPB is mentioned as one of the tools used to assess walking function, and walking function is listed as one of six assessments within the mobility domain of fall risk assessment. This suggests that while the SPPB is an important indicator for assessing fall risk, it may not be sufficient when used alone[10]. Therefore, when interpreting the results of this study, additional factors other than the SPPB (e.g., demographic characteristics[49] and fall risk factors such as muscle strength and neurological factors[50] need to be considered in addition to the SPPB. Future studies should also include more diverse populations, such as older adults living in nursing homes, and utilize data collection methods that reflect fall risk factors, such as sarcopenia and psycho-cognitive factors, to allow for more comprehensive analyses. Forth, psychological factors, such as fear of falling[51] and anxiety, which are known to influence the fall risk[48], were not considered. Future studies should incorporate assessments of fear of falling and anxiety to comprehensively analyze how these psychological factors interact with step width variability. Finally, gait analysis results obtained in experimental settings may not fully reflect daily life gait patterns, and simple tests may not provide sufficiently reliable measurements[52]. In this regard, using wireless wearable monitoring systems can provide long-term data, and better reflect real daily life gait patterns, increasing the accuracy of the collected data[53]. Therefore, future studies should aim to replicate these findings in larger, more diverse populations and use a longitudinal design to assess the temporal relationship between gait characteristics and fall risk. Despite these limitations, this study has several strengths. The greater discriminatory power of step width variability as walking speed increases highlights its potential as an early indicator of fall risk, and suggests the possibility that spatiotemporal gait variability may be used as a predictor of fall risk. Therefore, it provides a basis for further studies on spatiotemporal gait variability to predict fall risk.

## Conclusions

This study found that gait variables and variability play an important role in distinguishing between the fall-risk and non-fall-risk groups. In particular, the step width variability at 20% increased walking speed, compared to that at the PS, was a significant factor for predicting the fall risk. These findings suggest that treadmill-based gait evaluation can serve as a valuable tool for identifying individuals at higher risk of falls. Our results can offer valuable insights for the development of future fall risk prediction models.

## Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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## Author contributions
Park and Bae designed the research and contributed to conceptualization and methodology. Kim, Lim, and Bae led the research, managed human resources, and played key roles in data collection, formal analysis, resource management, and data curation. Kim, Lim, Park, and Bae participated in writing the manuscript. All authors have read and approved the final version of the manuscript. Bae supervised the research.

## Declarations

## Competing interests
The authors declare no competing interests.

## Additional information
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